

ABSTRACT

DNA Repair Mechanisms of Antarctic Marine Bacteria in Extreme Environments

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The Antarctic continent presents a myriad of extreme environmental conditions and a unique isolation from the impacts of human society. Organisms in this region face constant threats to their genetic material, with various agents capable of damaging the DNA molecule, including the potent ultraviolet radiation (UV) emanating from sunlight. Additionally, the harsh environmental factors, particularly the extremely low temperatures, further contribute to the challenge of maintaining DNA stability. As a result, Antarctica serves as a compelling setting for studying how organisms navigate the preservation of genome integrity in such harsh conditions. In this study, we aim to delve into the mechanisms employed by microorganisms to survive in the face of these extreme Antarctic environments. To achieve this, we have conducted genome sequencing of three Antarctic marine bacteria. Two of these bacteria were isolated from sediment samples at depths of 280 meters, while the third was retrieved from a depth of 1500 meters in the Bransfield Strait. Deeper sea conditions entail higher concentrations of CO₂ under elevated atmospheric pressure, adding an additional layer of environmental challenge for these bacteria. The DNA extracted from these isolates was sequenced using the Illumina platform. Subsequently, we performed de novo genome assembly utilizing *SPAdes v3.15.4*, and assessed assembly metrics with *QUAST v5.2.0*. Genome completeness was evaluated using *BUSCO v5.3.1*. Furthermore, we conducted annotation via the Prokaryotic Genome Annotation Pipeline (PGAP) and identified the species based on the 16S ribosomal RNA (rRNA) sequence, employing the software *GTDB-Tk 2.1.0*. Additionally, we scrutinized the presence of DNA repair genes in these organisms using blastp, with hits meeting criteria of e-value < 1E-5, coverage > 60%, and identity > 30% considered as orthologs. Hits not meeting these criteria had their conserved protein domains analyzed on InterPro to capture potential false-negatives. Preliminary findings indicate that the majority of DNA repair genes are indeed present in these bacteria, suggesting a diverse array of strategies they employ to combat DNA damage. Notably, nucleotide excision repair emerged as the most conserved pathway in these bacteria. These results underscore the significance of repair pathways involved in the removal of sunlight-induced lesions, indicating their crucial role in maintaining genome integrity in an environment characterized by high levels of UV radiation.

Keywords: Extreme Environments; Antarctica; DNA Repair.

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